Project Name:

LIBRARY MANAGEMENT SYSTEM.

Course Name:

Programming in Java.

Subject Code:

CSE2006.

Submitted By:

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Submitted To:

VIT BHOPAL University.

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INTRODUCTION:

This report details the design, implementation, and testing of the Library Management System. This project mainly uses the concept of Object-Oriented Programming, specifically Multilevel Inheritance, to model a real-world library environment. The system provides a functional simulation of standard library operations, offering a modular, efficient, and user-friendly experience.

Problem Statement:

Traditional library systems often require efficient software solutions to manage large volumes of books and users. The core problem addressed in this project is the design and implementation of a simplified, console-based library management system that automates essential operations. The system should manage book records, handle issue and return operations, and maintain data accuracy while ensuring ease of use.

Functional Requirements:

1. Book Management Module:

This system allows the user to add new books by entering details such as Book ID, Title, Author, Genre, and Availability Status.

•Input: User data via console.

•Output: A new Book object is created and stored in memory.

2. Issue and Return Module:

This system allows users to issue and return books. A book can only be issued if it is available.

•Input: Book ID and user action.

•Output: Updated availability status or an error message.

3. Information Display Module:

This system displays all book details, including availability status and issued records.

•Input: User menu selection.

•Output: Formatted book information.

Non-Functional Requirements:

1. **Usability:**

The system provides an intuitive, numbered console menu for easy navigation.

2. **Reliability:**

The system handles invalid inputs (e.g., wrong Book ID, invalid options) without crashing and displays appropriate error messages.

3. **Performance:**

Operations such as issuing and returning books execute efficiently and provide real-time feedback.

4. **Maintainability:**

The code is modular due to multilevel inheritance, with proper comments and meaningful variable names.

System Architecture:

The system follows a Layered Architecture implemented using Multilevel Inheritance.

- **Presentation Layer:** Handles user interaction through a menu-driven interface.
- **Business Logic Layer:** Contains logic for issuing, returning, and managing books.
- **Data Layer:** Stores book details and their current status.

Design Diagrams:
https://drive.google.com/file/d/19Npp3blndgQLnTY0BpkRXw-ihxp9q4Xm/view?usp=sharing

Design Decisions & Rationale:

- **Multilevel Inheritance:** Chosen to model a natural "is-a" relationship. A Library System is-a Library Operations which in turn is-a Book. This promotes code reusability and logical structuring of the system.
- **Encapsulation:** All instance variables are declared as protected to allow access to subclasses while restricting access from unrelated external classes.
- **Scanner Object:** Placed in the base class Book so that all derived classes can inherit and use the same Scanner object for input, avoiding resource duplication.

Implementation Details:

The system is implemented in a single .java file for simplicity, but the classes are logically separated. Key implementation aspects include:

- The Library System constructor initializes book details and manages system setup, demonstrating structured initialization.
- Input validation in issue() and return Book() methods ensures data integrity and prevents invalid operations.
- The library Menu() method uses a do-while loop to provide a continuous user experience.

Testing Approach:
A manual black-box testing approach was used.
- **Test Case 1 (Add Book):** Verified correct storage and display of book details.
- **Test Case 2 (Issue Book):** Issued a book successfully when available.
- **Test Case 3 (Issue Unavailable Book):** System displayed "Book not available!".
- **Test Case 4 (Return Book):** Book status updated correctly after return.

Challenges Faced:
- **Challenge 1:** Managing user input errors.
    - Solution: Implemented input validation and error handling.
- **Challenge 2:** Designing class hierarchy.
    - Solution: Separated responsibilities into different classes logically.

Learnings & Key Takeaways:
- Understood practical implementation of multilevel inheritance.
- Improved knowledge of Java OOP concepts.
- Learned handling of user input and system validation.

Future Enhancements:

• Implement a collection (like an ArrayList) to manage multiple books and users efficiently instead of handling limited records.
• Add a password-based authentication system for secure access to the library system.
• Replace the console interface with a Graphical User Interface (GUI) using Swing or JavaFX.
• Integrate with a database (e.g., MySQL) for persistent storage of book and user data.
• Implement a book reservation system and fine calculation for overdue returns.

References:

• Oracle Java
Documentation: https://docs.oracle.com/javase/8/docs/technotes/guides/language/index.html
• GeeksforGeeks - Multilevel Inheritance in Java.

Results Screenshots :

```
princeraj@princes-MacBook-Air-2 Library managenment system %  /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin
.plugin/Contents/Home/bin/java -cp /Users/princeraj/Library/Application\ Support/Code/User/workspaceStorage/58634291b4e
ce788b1631a7b624d8f67/redhat.java/jdt_ws/Library\ managenment\ system_5a8edec0/bin Main
No previous data found.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 1
Enter title: THE GREAT JAVA
Enter author: PRINCE RAJ
Book added successfully!

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 1
Enter title: BASIC JAV
Enter author: G ganeshan
Book added successfully!

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 2
0. THE GREAT JAVA | PRINCE RAJ | Issued: false
1. BASIC JAV | G ganeshan | Issued: false

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
```

Library managenment system

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: Main

```
princeraj@princes-MacBook-Air-2 Library managenment system %  /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin
.plugin/Contents/Home/bin/java -cp /Users/princeraj/Library/Application\ Support/Code/User/workspaceStorage/58634291b4e
ce788b1631a7b624d8f67/redhat.java/jdt_ws/Library\ managenment\ system_5a8edec0/bin Main
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 4
0. THE GREAT JAVA | PRINCE RAJ | Issued: false
1. BASIC JAV | G ganeshan | Issued: false
Enter index: 0
Book issued.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 4
0. THE GREAT JAVA | PRINCE RAJ | Issued: true
1. BASIC JAV | G ganeshan | Issued: false
Enter index: 1
Book issued.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 4
0. THE GREAT JAVA | PRINCE RAJ | Issued: true
1. BASIC JAV | G ganeshan | Issued: true
Enter index: 1
Book already issued.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 5
```

0 0 Java: Ready Ln 13, Col 58 Spaces: 4 UTF-8 LF { } Java Go Live

Library managenment system

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: Main

```
princeraj@princes-MacBook-Air-2 Library managenment system %  /usr/bin/env /Library/Internet\ Plug-Ins/JavaAppletPlugin
.plugin/Contents/Home/bin/java -cp /Users/princeraj/Library/Application\ Support/Code/User/workspaceStorage/58634291b4e
ce788b1631a7b624d8f67/redhat.java/jdt_ws/Library\ managenment\ system_5a8edec0/bin Main
6. Exit
Enter choice: 4
0. THE GREAT JAVA | PRINCE RAJ | Issued: true
1. BASIC JAV | G ganeshan | Issued: false
Enter index: 1
Book issued.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 4
0. THE GREAT JAVA | PRINCE RAJ | Issued: true
1. BASIC JAV | G ganeshan | Issued: true
Enter index: 1
Book already issued.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 5
0. THE GREAT JAVA | PRINCE RAJ | Issued: true
1. BASIC JAV | G ganeshan | Issued: true
Enter index: 1
Book returned.

--- Library Menu ---
1. Add Book
2. View Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter choice: 6
Error saving file.
Data saved. Exiting...
princeraj@princes-MacBook-Air-2 Library managenment system %
```

0 0 Java: Ready Ln 13, Col 58 Spaces: 4 UTF-8 LF { } Java Go Live